

Part Two: Factor Analysis and Structural Equation Modeling

Jinsong Chen

Table of contents

1	What Part Two Is	1
2	The Measurement Problem at a Glance	2
3	How Chapters 1–4 Progress	3
4	One Model Family, Several Responsibilities	5
5	Evidence Roles Across the Sequence	6
6	Boundaries and What Follows	7
7	Routes and Commitments	7

1 What Part Two Is

Part Two of *Quantitative Methods for Social and Behavioral Research with R* introduces latent-variable modeling through factor analysis and then prepares the transition to structural equation modeling. It is written for postgraduate students and applied researchers who need to understand what a latent-variable model asserts, how its parameters are estimated, and how far its results can support a substantive interpretation.

The first four chapters form one measurement-model sequence. Chapter 1 introduces the common-factor model and its representations. Chapter 2 develops an exploratory factor model. Chapter 3 turns the resulting proposal into a restricted confirmatory model and estimates it. Chapter 4 evaluates that fit, compares planned rivals, studies a frozen revision, fits higher-order and bifactor representations, and closes with a retrospective known-population audit. The sequence is cumulative: later chapters assume the model, notation, and analytic history established earlier.

This is not a catalogue of software commands or fit-index cutoffs. Equations, matrices, diagrams, prose, and code are treated as different representations of the same model. Software makes the calculations reproducible; it does not choose the research question, justify a construct label, or turn an exploratory decision into an independently confirmed result.

What is assumed. Readers should understand variance, covariance, correlation, sampling uncertainty, and elementary regression notation at the level developed in Part One. Basic matrix multiplication is helpful, but the dimensions and operations needed for each model are explained where they are used. Calculus and proof-based mathematical statistics are not prerequisites.

How to use this map. This introduction explains why Chapters 1–4 appear in their present order, how their evidence roles differ, and what they hand to the later structural-modeling sequence. It is an orientation document, not a substitute for any chapter.

2 The Measurement Problem at a Glance

Several observed indicators may vary together even though the attribute used to explain their association is not recorded directly. A common-factor model represents that proposal as

$$\mathbf{y}_i = \nu + \Lambda\eta_i + \epsilon_i, \quad \begin{aligned} E(\mathbf{y}_i) &= \nu + \Lambda\alpha, \\ \Sigma(\vartheta) &= \Lambda\Phi\Lambda^\top + \Theta. \end{aligned}$$

Here $\mathbf{y}_i$ contains observed indicators, η_i contains unobserved factors, Λ contains loadings, and ϵ_i contains what the declared factors leave unexplained. The vectors ν and α are indicator and latent intercepts; with no structural regressions in this measurement-only model, $\alpha = E(\eta_i)$. Under $E(\epsilon_i) = \mathbf{0}$ and $\text{Cov}(\eta_i, \epsilon_i) = \mathbf{0}$, $\Phi = \text{Var}(\eta_i)$ and $\Theta = \text{Var}(\epsilon_i)$ produce the displayed moments; ϑ collects the model's free parameters. The covariance equation divides observed variation into a common part and a residual part. It also gives the model observable consequences: proposed loadings, factor relations, and residual restrictions imply a particular covariance matrix.

The equation does not make a factor identical to the construct used to name it. A factor is an unobserved random variable within a statistical model. A construct is a substantive concept whose interpretation also depends on the indicator domain, response process, population, design, and intended use. Likewise, a factor-to-indicator arrow states the direction of representation; by itself it does not establish that the factor causally produces a response. Good covariance fit can support a statistical representation without proving that a latent entity exists exactly as named.

Part One's question–answer discipline remains in the background: a method should answer a declared question, and the conclusion should not outrun its evidence and justification. Part Two does not repeat the complete argument-based framework. Chapter 1 contains one optional

late application to the latent-variable context; Chapters 2–4 concentrate on the statistical work needed to develop, estimate, and evaluate measurement models.

3 How Chapters 1–4 Progress

The four chapters make four connected moves:

1. **Represent:** state what factors and residuals mean and derive the moments implied by a common-factor model.
2. **Explore and develop:** use a development record to investigate dimensionality, orientation, interpretation, and sensitivity, then freeze a candidate.
3. **Specify and estimate:** express that candidate as explicit restrictions, establish recoverability, fit it to unused data, and audit the solution.
4. **Evaluate, compare, and extend:** assess global and local discrepancy, compare eligible rivals, study revisions honestly, and examine advanced measurement representations.

Figure 1 makes the architecture explicit. Each stage changes the question and evidence role while retaining the same common-factor foundation.

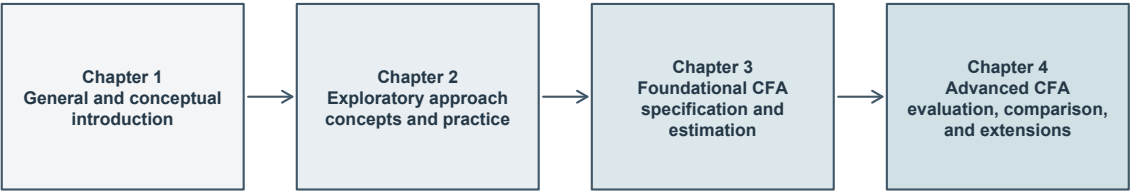


Figure 1: The four-chapter architecture moves from representation to exploratory development, foundational CFA, and advanced CFA.

Table 1 summarizes the responsibility and handoff of each chapter.

Table 1

The four-chapter measurement-model sequence.

Chapter	Main responsibility	What it hands forward
1. From Latent-Variable Modeling to Factor Analysis	Distinguish observed variables, factors, constructs, composites, and scores; translate the common-factor model among scalar equations, matrices, diagrams, and prose; derive implied means and covariances; explain residuals and latent-coordinate conventions	A technically explicit common-factor representation, known-value teaching records, and the questions that exploratory analysis must answer
2. Exploratory Factor Analysis and Model Development	Audit indicators and their correlations; choose an extraction target; investigate factor count; rotate and interpret neighboring solutions; examine residuals, sensitivity, and stability	A documented three-factor candidate, named alternatives, and unresolved questions frozen before unused confirmation data are inspected
3. Specifying and Estimating Confirmatory Factor Models	Translate the candidate into fixed, free, and constrained parameters; set factor metrics; count moments and coordinates; examine identification with small-model diagrams; translate a published four-factor model into syntax and a parameter ledger; estimate by continuous ML; read convergence, admissibility, coefficients, and uncertainty	A reproducible fitted object and parameter ledger whose overall adequacy has deliberately not yet been decided
4. Evaluating and Comparing Confirmatory Factor Models	Read global fit as a profile; locate local discrepancy; compare compatible planned rivals; interpret modification diagnostics; freeze and revalidate a revision; fit and compare higher-order and bifactor representations; conduct a retrospective known-population audit; examine reliability and score boundaries	A calibrated measurement-model conclusion and a bridge from relations among factors to later structural equation models

The stopping points are deliberate. Chapter 2 does not relabel a developed pattern as

confirmation. Chapter 3 does not treat convergence as adequate fit. Chapter 4 does not treat favorable fit as construct validation. Each chapter completes one responsibility before transferring a clearly defined record to the next.

The diagrams also form a deliberate progression. Chapter 3 reads the same visual grammar across a two-factor algebra model, the three-factor SLE model, and a published four-factor transfer, then uses small paired diagrams to make identification and equivalence visible. Chapter 4 carries that grammar into the three-factor higher-order and bifactor alternatives. The diagrams, syntax, and parameter counts are generated from matching structural ledgers rather than treated as independent illustrations.

4 One Model Family, Several Responsibilities

The common-factor covariance equation remains the mathematical backbone of all four chapters. What changes is the analytical responsibility assigned to it. Chapter 1 works forward from known parameters to implied moments. Chapter 2 works from sample associations toward a provisional low-dimensional representation. Chapter 3 asks whether a prespecified parameterization can be recovered and estimated. Chapter 4 asks how much discrepancy remains, where it is located, and which comparisons are warranted.

Four mathematical threads connect the sequence:

- **Common and residual variation.** Scalar covariance expansions become the matrix decomposition $\Lambda\Phi\Lambda^\top + \Theta$, then become fitted residuals and local diagnostics.
- **Coordinates and restrictions.** Scale, sign, and rotational freedom lead to EFA rotation, CFA scaling constraints, identification checks, and the distinction between equivalent and testably different models.
- **Information and estimation.** Sample means and covariances become targets for a restricted model; the number of distinct moments, free coordinates, and candidate degrees of freedom is counted consistently before numerical output is interpreted.
- **Evaluation and use.** Parameter uncertainty, global fit, local residuals, planned comparisons, reliability, and scores answer different questions. None is a universal pass–fail certificate for the model or construct.

The statistical level is intentionally higher than an introductory survey. Core derivations remain in the chapters and are explained in words rather than removed. The lecture decks stage the same ideas across shorter teaching steps; exercises require calculations and interpretations; and the R labs make selected analyses executable without turning the chapter into a command reference.

Chapter 1’s main route requires no student software. Chapter 2 introduces the `psych` and `GPArotation` workflow for EFA. Chapters 3–4 use `lavaan` for CFA estimation and evaluation. In every case the printed chapter supplies enough information to understand the model and the conclusion even when the reader does not run the code.

5 Evidence Roles Across the Sequence

Repeated use of one topic does not mean that every record has the same inferential role. Part Two makes the analysis history visible so that development, first evaluation, and revalidation are not collapsed into one favorable story. Table 2 gives the principal roles.

Table 2

Records keep distinct analytical roles even when they illustrate the same model family.

Record	First permitted use	What it cannot establish by itself
Known-value six-indicator records	Chapter 1 reconstructs covariance from declared parameters	Sampling behavior, empirical factor count, or the existence of named constructs
SLE Sample A, 360 cases	Chapter 2 develops and freezes the exploratory candidate	Independent confirmation of choices made while examining A
SLE Sample B, 360 cases	Chapter 3 estimates the frozen candidate; Chapter 4 evaluates it and planned rivals	Independent support for a new relation first suggested by B
SLE Sample C, 360 cases	Chapter 4 evaluates one revision frozen after B	General validity, invariance, or permission to search repeatedly in C
Known SLE generating moments	Chapter 4 reveals them retrospectively, after the B/C analysis history is complete	Contemporaneous justification for earlier choices, or finite-sample bias inferred from one realization
Separate donor microcases	Compact demonstrations of specified calculations and model comparisons	The development or confirmation status of the SLE sequence

The Student Learning Experiences (SLE) records are synthetic continuous data, not observations from actual students. One file, `sle_continuous.csv`, stores 1,080 rows across three independent samples of 360 cases and twelve indicators: four proposed Planning indicators, four Strategy indicators, and four Belonging indicators. A `sample` column identifies A, B, or C, and the Chapter 2 appendix supplies the codebook and manifest. The samples are selected for their declared roles, never pooled to manufacture a larger analysis. A record that becomes exploratory through revision does not recover confirmatory status merely because the next command calls the model CFA.

Two numerical roles are kept separate in Chapter 4. Its extension fits use the factor correlations estimated from Sample B. Only afterward does the retrospective audit reveal the generating correlation triplet (.42, .30, .35) and its exact higher-order coordinates (.60, .70, .50). Those

known values sharpen the teaching comparison; they are not Sample-B estimates and were not evidence used to develop or revise the candidate.

This separation is also a teaching device. It lets readers see that a model can be technically coherent, reproducible, and even apparently well fitting while still owing a narrower conclusion than its name might suggest.

6 Boundaries and What Follows

The initial Part Two sequence treats indicators as continuous and keeps one coherent continuous-ML lineage. Ordered categorical indicators, thresholds, polychoric correlations, and their estimator-specific testing conventions are signposted but taught elsewhere. This boundary does not exclude categorical grouping variables or coded predictors: later multigroup, invariance, and structural models may use them while retaining continuous indicators.

The four-chapter architecture focuses on reflective common-factor measurement models. It does not make a factor score an error-free observation, equate fit with validity, or establish causality from a path diagram. Within that architecture, Chapter 4 treats higher-order and bifactor models as rival measurement representations, not automatic remedies for a weak core model. Specialized topics such as item-response theory, mixture models, multilevel factor analysis, and a full treatment of categorical indicators belong to other courses or later series.

After Chapter 4, Part Two turns from relations between factors and indicators to relations among variables in a structural system. Observed-variable path analysis prepares directed systems; general structural equation modeling combines structural relations with measurement models; measurement invariance supports comparisons across groups or occasions; and latent growth models represent both average trajectories and variation in change. Multilevel growth belongs to Part Three.

The notation is chosen to survive that transition. In particular, a latent intercept is not generally the same as the marginal latent mean once structural regressions are introduced. The later chapters extend the model rather than quietly changing what its symbols mean.

7 Routes and Commitments

A taught course should follow Chapters 1–4 in order. The lecture for each chapter covers most of its conceptual and mathematical route; the exercise companion supplies worked reference answers; and the R lab offers in-class practice. In Chapters 3–4 these companions carry the same diagram-reading, extension, and audit distinctions as the public chapters while keeping R-lab datasets in their separate practice role. Homework extends the same responsibilities. Part Two quizzes are not yet part of the teaching package.

A reader working alone should begin with Chapter 1, then keep Chapters 2–4 in sequence so that the exploratory candidate, confirmatory restrictions, fitted object, and evaluation history remain intelligible. The labs may be run alongside the chapters but are not prerequisites for the mathematical route.

A reader consulting one method should read the chapter’s opening question, record and destination before jumping to an output table. A fit index, loading, modification index, or reliability estimate has meaning only within the model, metric, evidence role, and intended use that produced it.

Across these routes, six commitments remain constant:

1. **The model agrees across representations.** Prose, equations, matrices, diagrams, syntax, and output must describe the same restrictions.
2. **Exploration and confirmation are historical roles.** They depend on when a decision was made and which evidence informed it, not on a function name.
3. **A fitted solution is read in stages.** Identification, numerical convergence, admissibility, uncertainty, global fit, and local fit are connected but not interchangeable checks.
4. **Comparison requires eligibility.** Nested tests and information criteria apply only when their assumptions and data-likelihood bases make the comparison meaningful.
5. **Computation reproduces rather than justifies.** Code can verify arithmetic and preserve an analysis record; it cannot supply missing design, measurement, or substantive warrants.
6. **Claims remain bounded.** Covariance compatibility does not by itself prove dimensionality, construct reality, validity, invariance, score fitness, or causal direction.

Each numbered chapter closes with a **Summary and reading bridge**, followed by unnumbered exercises. Appendices may add derivations or software details, but the principal question and its calibrated answer remain understandable from the numbered route.